

digi-soul

Nervous System & Brain-Organ Coupling

Research Basis for the digi-soul NervousSystem Module

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Repository: /Users/mehmetkucuk/Documents/GitHub/digi-soul

Implementation: PR #1, merge SHA 04521d5 (origin/main)

This report documents the peer-reviewed literature, physiological constants, and citation evidence that grounded the recent digi-soul NervousSystem upgrade. Research by dsg-bio-researcher; report compiled by dsg2.

1. Key Findings

1. CNS/PNS architecture & brain interface

- The PNS divides into somatic (voluntary) and autonomic (involuntary) branches; the autonomic branch is the primary brain-to-organ control channel.
- The brainstem (medulla, pons), hypothalamus, and insular/prefrontal cortex form the central autonomic network.
- Rapid signaling depends on myelination / saltatory conduction (Draghici & Taylor 2018, PMID 28844537; Salzer 2015, PMID 26054742).

2. Autonomic organ regulation

- The arterial baroreflex is the canonical fast negative-feedback loop; vagal cardiac control often persists when sympathetic pathways are disrupted.
- Sympathetic outflow is graded (rate + recruitment + temporal coding), not on/off (Shoemaker 2017, PMID 28871339).

3. Neural signaling

- Action-potential propagation speed is set by axon diameter and myelination (Salzer 2015, PMID 26054742).

4. Brain-organ feedback

- Three parallel bidirectional channels - neural (vagus/enteric), endocrine (HPA axis), immune (cytokines) (Asadi 2022, PMID 35421277).
- HPA feedback can break down: CRH/ACTH 'inappropriately normal' despite high cortisol; a proposed CRH-norepinephrine positive-feedback loop drives hypercortisolism (Perrelli 2024, PMID 38927393).
- HRV and baroreflex sensitivity are validated non-invasive readouts of vagal/sympathetic balance. In Guillain-Barre syndrome, LF-HRV ($p=0.027$), HF-HRV ($p=0.008$), total power ($p=0.015$), and BRS slopes ($p=0.034/0.011$) are all significantly reduced (Tan 2019, PMID 29654380).

Animal-data flag: Huzard 2019 (PMID 30448728) is rodent-only - treated as a mechanistic hypothesis only, not a human parameter source.

Citation counts via Semantic Scholar, retrieved 2026-06-28.

Page 3

3. Physiological Constants (model-ready)

Conduction velocities (Kandel)

Aa 80-120 m/s
Ab 35-75 m/s
Ad 5-30 m/s
B (preganglionic autonomic) 3-15 m/s
C (unmyelinated) 0.5-2 m/s

Membrane / action potential

resting -70 mV; threshold -55 mV; spike peak +30 to +40 mV
AP duration 1-2 ms; refractory 1-2 ms; synaptic delay 0.5-1 ms

Autonomic latency asymmetry

vagal effect fast (<1 s, beat-to-beat)
sympathetic slow (onset 1-5 s, peak 20-30 s)

HRV bands (Task Force 1996) & norms (Shaffer & Ginsberg 2017)

VLf <0.04 Hz; LF 0.04-0.15 Hz; HF 0.15-0.40 Hz
Healthy norms: SDNN ~50 ms, RMSSD ~42 ms, LF/HF ~1.5-2.0

Baroreflex & heart rate

Baroreflex sensitivity ~10-20 ms/mmHg (healthy adults)
Intrinsic HR ~100-110 bpm (full autonomic blockade)
Resting HR 60-80 bpm reflects net vagal dominance

4. Implementation Outcome (what shipped)

Merged to main as PR #1 (merge SHA 04521d5). Five files touched:

- nervous_system/autonomic.py
- nervous_system/nervous_system.py
- organs/vascular_system.py
- circuit.py
- canvas/display.py

Delivered:

1. Dual autonomic effector pathways with vagal-fast / sympathetic-slow latency asymmetry.
2. Graded saturating sympathetic output (rate + recruitment), not on/off.
3. Baroreflex negative-feedback loop with emergent HRV diagnostics.
4. Per-fiber conduction-velocity delays.
5. Tunable HPA cortisol->CRH/ACTH feedback gain; detuning reproduces the Perrelli 2024 pathology.
6. GUI rendering of autonomic / HRV / HPA diagnostics in the Neural tab.

Two bugs fixed during integration:

- a missing _autonomic_loop() AttributeError
- a baroreflex resting set-point bug

5. Known Gaps / Follow-up

- Semantic Scholar citation counts for Draghici 2018 and Asadi 2022 are still pending (API rate-limited; retry after cooldown or with an x-api-key token).
- Brain-to-liver/stomach/kidney autonomic latencies are under-quantified in humans - parameterized by fiber type rather than measured constants.

Suggested follow-up research:

- age-stratified HRV norms
- vagal control of gastric secretion
- renal sympathetic / ADH coupling
- hepatic sympathetic glucose output
- cholinergic anti-inflammatory pathway (Tracey)